

Assignment 1

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Discrete Systems

1 The system S is given by $y(n) = y(n-1) - x(n-1) + x(n)$. Assume that $y(-1) = 0$

1.a Compute the output to the input $x(n) = \delta(n)$

n	0	1	2	3	4	5
$x(n)$	1	0	0	0	0	0
$y(n)$	1	0	0	0	0	0

1.b Compute the output to the input $x(n) = \delta(n) + \delta(n-1) + \delta(n-2)$

n	0	1	2	3	4	5
$x(n)$	1	1	1	0	0	0
$y(n)$	1	1	1	0	0	0

1.c Compute the output to the input $x(n) = \cos(\pi n)$

n	0	1	2	3	4	5
$x(n)$	1	-1	1	-1	1	-1
$y(n)$	2	0	2	0	2	0

Difference Equations to State Space Equations

2 Given the difference equation $y(n) = y(n-1) - y(n-2) + x(n) - x(n-2)$, determine the (A, B, C, D) representation of the system.

2.I Find $\overrightarrow{S(n)}$

Let:

$$s_1(n) = x(n-2)$$

$$s_2(n) = x(n-1)$$

$$s_3(n) = y(n-2)$$

$$s_4(n) = y(n-1)$$

Then:

$$s_1(n+1) = x(n-1) = s_2(n)$$

$$s_2(n+1) = x(n)$$

$$s_3(n+1) = y(n-1) = s_4(n)$$

$$s_4(n+1) = y(n)$$

$$= y(n-1) - y(n-2) + x(n) - x(n-2)$$

$$= -s_1(n) - s_3(n) + s_4(n) + x(n)$$

Thus, let $\overrightarrow{S(n)} = \begin{pmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \end{pmatrix}$

2.II Find $\overrightarrow{S(n+1)}$

$$\begin{aligned} \overrightarrow{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \\ s_3(n+1) \\ s_4(n+1) \end{pmatrix} \\ &= \begin{pmatrix} s_2(n) \\ x(n) \\ s_4(n) \\ -s_1(n) - s_3(n) + s_4(n) + x(n) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} x(n) \end{aligned}$$

Therefore, our $\mathbf{A}$ and $\vec{B}$ matrices are $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$.

2.III Find $y(n)$

$$\begin{aligned}
 y(n) &= -x(n-2) - y(n-2) + y(n+1) + x(n) \\
 &= -s_1(n) - s_3(n) + s_4(n) + x(n) \\
 &= \begin{pmatrix} -1 & 0 & -1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)
 \end{aligned}$$

Therefore, our $\mathbf{C}$ and $\mathbf{D}$ matrices are $\begin{pmatrix} -1 & 0 & -1 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.
 Putting it all together, our system can be represented in state space form as:

$$\overrightarrow{S(n+1)} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} x(n)$$

$$y(n) = \begin{pmatrix} -1 & 0 & -1 & 1 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)$$

(A, B, C, D) Representation

3 Given the system $y(n) = x(n) + x(n-2) - y(n-2)$

3.a Find the (A, B, C, D) representation of the system

3.a.I Find $\overrightarrow{S(n)}$

Let:

$$s_1(n) = x(n-2)$$

$$s_2(n) = x(n-1)$$

$$s_3(n) = y(n-2)$$

$$s_4(n) = y(n-1)$$

Then:

$$s_1(n+1) = x(n-1) = s_2(n)$$

$$s_2(n+1) = x(n)$$

$$s_3(n+1) = y(n-1) = s_4(n)$$

$$\begin{aligned} s_4(n+1) &= y(n) \\ &= -y(n-2) + x(n-2) + x(n) \\ &= s_1(n) - s_3(n) + x(n) \end{aligned}$$

Thus, let $\overrightarrow{S(n)} = \begin{pmatrix} s_1(n) \\ s_2(n) \\ s_3(n) \\ s_4(n) \end{pmatrix}$.

3.a.II Find *overrightharpoon* $\overrightarrow{S(n+1)}$

$$\begin{aligned} \overrightarrow{S(n+1)} &= \begin{pmatrix} s_1(n+1) \\ s_2(n+1) \\ s_3(n+1) \\ s_4(n+1) \end{pmatrix} \\ &= \begin{pmatrix} s_2(n) \\ x(n) \\ s_4(n) \\ s_1(n) - s_3(n) + x(n) \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$$

Therefore, our $\mathbf{A}$ and $\vec{B}$ matrices are $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$ respectively.

3.a.III Find $y(n)$

$$\begin{aligned}
 y(n) &= -y(n-2) + x(n-2) + x(n) \\
 &= s_1(n) - s_3(n) + x(n) \\
 &= \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)
 \end{aligned}$$

Therefore, our $\mathbf{C}$ and $\mathbf{D}$ matrices are $\begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \end{pmatrix}$ respectively.
 Putting it all together, our system can be represented in state space form as:

$$\begin{aligned}
 \overrightarrow{S(n+1)} &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \\
 y(n) &= \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix} \overrightarrow{S(n)} + \begin{pmatrix} 1 \end{pmatrix} x(n)
 \end{aligned}$$

3.b Use the state space equations to compute 5 outputs to the input $x(n) = \delta(n)$

Current State ($\vec{S}(n)$)

Current Output ($y(n)$)

$$\vec{S}(0) = \vec{0}$$

$$y(0) = \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix} \vec{S}(0) + \begin{pmatrix} 1 \end{pmatrix} x(0) = 1$$

$$\vec{S}(1) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \vec{S}(0) + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} x(0) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

$$y(1) = \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix} \vec{S}(1) + \begin{pmatrix} 1 \end{pmatrix} x(1) = 0$$

$$\vec{S}(2) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \vec{S}(1) + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} x(1) = \vec{0}$$

$$y(2) = \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix} \vec{S}(2) + \begin{pmatrix} 1 \end{pmatrix} x(2) = 0$$

$$\vec{S}(3) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \vec{S}(2) + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} x(2) = \vec{0}$$

$$y(3) = \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix} \vec{S}(3) + \begin{pmatrix} 1 \end{pmatrix} x(3) = 0$$

$$\vec{S}(4) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \vec{S}(3) + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} x(3) = \vec{0}$$

$$y(4) = \begin{pmatrix} 1 & 0 & -1 & 0 \end{pmatrix} \vec{S}(4) + \begin{pmatrix} 1 \end{pmatrix} x(4) = 0$$

Compound Interest

- 4 **A bank account is just a system:** Letting $x(n)$ represent the amount deposited or withdrawn (with $x(n) > 0$ representing deposits and $x(n) < 0$ representing withdrawals), and α represent the interest rate, we can express the account balance with the system $y(n) = y(n-1) + \alpha y(n-1) + x(n)$. How much money do you have after 10 days if you initially deposit \$100?

We're just going to solve this through direct computation with a relatively simply inference about $y(n)$:

n	0	1	2	3	...	10
$x(n)$	100	0	0	0	...	0
$y(n)$	100	$100(1 + \alpha)$	$100(1 + \alpha)^2$	$100(1 + \alpha)^3$	...	$100(1 + \alpha)^{10}$

Therefore, with an initial deposit of \$100, the account balance grows to $\$100(1 + \alpha)^{10}$ after 10 days.